

1. Test the search feature.

What's missing: No product type (e-commerce, hospital),

no filter/sort behavior, no empty/invalid input scenarios, no performance angle.

Test the search feature of amazon website

Amazon's search feature is a comprehensive discovery tool powered by advanced algorithms (like the A9 engine) and modern AI capabilities. It goes far beyond a simple keyword text box to help users pinpoint exact items.

Key components of Amazon's search capabilities include:

- **Predictive Autocomplete:** As you type in the search bar, it predicts what you are looking for based on popular trends and past searches.
- **Advanced Filters & Sorting:** You can sort and filter results by department, price range, brand, customer ratings, Prime eligibility, and specific product attributes (like size, color, or material).
- **Visual Search & AI Features:** Tools like **Amazon Lens** allow users to snap photos or upload images to find visually similar items. Amazon also incorporates real-time AI image generation as you type descriptive queries (such as textures or patterns) to help visualize and discover products in categories like apparel and home goods.
- **Boolean & Special Operators:** Users can refine text queries using operators like quotation marks for exact phrases, minus signs to exclude terms, and wildcards (*k*) for word variations.

If you'd like, I can help you search for specific products, compare options across different categories, or find deals right now. Would you like me to look up a particular item for you?

2. "Generate test cases for file upload."

No file types, no size limits, no success/failure criteria, no security angle (malicious file upload).

Here is a comprehensive set of test cases for a standard **File Upload** feature, categorized by test type to ensure thorough coverage:

1. Positive Test Cases (Happy Path)

- **TC-POS-01: Valid File Format (Single)**

- *Description:* Upload a single file with a supported format (e.g., .jpg, .png, .pdf) and within the allowed size limit.
- *Expected Result:* File uploads successfully, displays a success message, and shows a preview or file name.

- **TC-POS-02: Multiple File Upload (if supported)**

- *Description:* Select and upload multiple valid files simultaneously up to the maximum allowed count.
- *Expected Result:* All files upload successfully and are listed correctly in the UI.

- **TC-POS-03: Drag and Drop Functionality**

- *Description:* Drag a valid file from the local file system and drop it into the designated drop zone.
- *Expected Result:* The file is accepted and processed for upload just like using the file browser.

2. Negative Test Cases (Validation & Errors)

- **TC-NEG-01: Invalid File Extension**

- *Description:* Attempt to upload a file with an unsupported extension (e.g., .exe, .bat, .sh on a portal that only accepts images).
- *Expected Result:* The upload is rejected immediately with a clear error message (e.g., "Invalid file type. Allowed formats: JPG, PNG, PDF").

- **TC-NEG-02: File Size Exceeds Limit**

- *Description:* Upload a valid file type whose size is strictly greater than the maximum permitted limit (e.g., uploading a 15MB file when the limit is 10MB).
- *Expected Result:* The system blocks the upload and displays an error message stating the maximum size limit.

- **TC-NEG-03: Zero-Byte (Empty) File**

- *Description:* Attempt to upload an empty file (0 KB).
- *Expected Result:* The system rejects the file with an appropriate error message.

- **TC-NEG-04: Exceeding Maximum File Count**

- *Description:* Try to select more files than the maximum allowed limit in a single batch.
- *Expected Result:* The system either restricts selection or shows an error indicating the maximum number of files allowed.

3. Boundary Value Analysis

- **TC-BND-01: Exact Maximum File Size**

- *Description:* Upload a file that is precisely at the maximum allowed size limit (e.g., exactly 10.0 MB).
- *Expected Result:* The file uploads successfully.

- **TC-BND-02: Maximum File Size + 1 Byte**

- *Description:* Upload a file that exceeds the maximum size limit by just 1 byte.
- *Expected Result:* The file is rejected.

- **TC-BND-03: Maximum Filename Length**

- *Description:* Upload a file with an extremely long filename (e.g., 255+ characters).
- *Expected Result:* The system either successfully handles the long name, truncates it gracefully, or displays a validation error without crashing.

- **TC-BND-04: Special Characters in Filename**

- *Description:* Upload a file with special characters, spaces, or international/Unicode characters in the filename (e.g., test @file #1.pdf).
- *Expected Result:* The system successfully sanitizes the filename and uploads the file without throwing an error.

4. Security Test Cases

- **TC-SEC-01: Malicious File Upload (EICAR / Antivirus Test)**
 - *Description:* Attempt to upload a harmless test virus file (like the EICAR standard anti-malware test file) or a script disguised as a valid format.
 - *Expected Result:* The server-side security scan detects the threat and blocks the upload.
- **TC-SEC-02: Extension Spoofing / Double Extension**
 - *Description:* Upload an executable file renamed with a fake extension (e.g., malware.php.jpg).
 - *Expected Result:* The backend performs deep content inspection (MIME type / magic bytes check) rather than relying solely on the file extension, and rejects the file.
- **TC-SEC-03: Path Traversal Attack**
 - *Description:* Upload a file with a manipulated filename containing path traversal characters (e.g., ../../etc/passwd.jpg).
 - *Expected Result:* The application sanitizes the filename to prevent writing outside the designated upload directory.

5. Performance & Usability Test Cases

- **TC-PER-01: Progress Indicator**

- *Description:* Upload a large valid file over a throttled network connection.
- *Expected Result:* A progress bar or loading spinner is displayed accurately indicating the upload progress.
- **TC-PER-02: Cancel Ongoing Upload**
 - *Description:* Initiate an upload for a large file and click the "Cancel" button mid-way.
 - *Expected Result:* The upload stops immediately, and any partially uploaded data is purged from the server.
- **TC-PER-03: Concurrent Uploads**
 - *Description:* Multiple users attempt to upload files simultaneously under peak load conditions.
 - *Expected Result:* The server handles the requests without timing out or crashing.

With ICEPOT reference; "Generate test cases for file upload of pdf files of not more than 5 MB including malicious files"

Here is a targeted set of test cases designed specifically for uploading **PDF files** with a **5 MB size limit**, including security and malicious file scenarios:

1. Positive Test Cases (Valid PDF Uploads)

- **TC-POS-01: Standard PDF Upload**

- *Description:* Upload a valid PDF file that is well under the limit (e.g., a 2 MB text-based PDF).
 - *Expected Result:* The file uploads successfully, displays a success confirmation, and provides a correct preview or file name.
- **TC-POS-02: Boundary Value - Exact Maximum Size (5 MB)**

- *Description:* Upload a valid PDF file whose size is precisely 5.0 MB.
 - *Expected Result:* The file uploads successfully without any error messages.

2. Negative Test Cases (Size & Format Validation)

- **TC-NEG-01: File Size Exceeds Limit (5.01 MB)**

- *Description:* Attempt to upload a valid PDF file that is just over the limit (e.g., 5.1 MB).
 - *Expected Result:* The upload is blocked, and an error message is displayed stating that the file exceeds the maximum allowed size of 5 MB.

- **TC-NEG-02: Large File Size**

- *Description:* Attempt to upload a large PDF (e.g., 25 MB).
 - *Expected Result:* The system immediately rejects the file or stops the upload, displaying a size limit error.

- **TC-NEG-03: Invalid File Format**

- *Description:* Attempt to upload a non-PDF file (e.g., .docx, .jpg, .txt, or .exe) within the 5 MB limit.
 - *Expected Result:* The system rejects the file and displays an error message indicating that only PDF files are allowed.
- **TC-NEG-04: Corrupted or Malformed PDF**
- *Description:* Upload a file with a .pdf extension, but the internal file structure is broken or corrupted.
 - *Expected Result:* The system rejects the file during processing and shows an error stating the file is unreadable or corrupted.

3. Security & Malicious File Test Cases

- **TC-SEC-01: Extension Spoofing (Disguised Executable)**
- *Description:* Take a malicious executable or script (e.g., .exe or .bat), rename the extension to .pdf (e.g., invoice.pdf), and attempt to upload it.
 - *Expected Result:* The backend performs deep content inspection (checking the file's "magic bytes" or MIME type rather than trusting the extension alone), identifies it as a non-PDF format, and blocks the upload.
- **TC-SEC-02: PDF Containing Embedded Malicious JavaScript/Exploit**

- *Description:* Upload a legitimate-looking PDF file that contains embedded malicious JavaScript or a known PDF exploit payload (often used for drive-by downloads or remote code execution).
 - *Expected Result:* The server-side security scanner or integrated antivirus/sandbox software detects the threat signature within the PDF structure and blocks the upload.
- **TC-SEC-03: Double Extension Attack**

- *Description:* Attempt to upload a file with a double extension (e.g., document.pdf.exe or document.exe.pdf).
- *Expected Result:* The system sanitizes and rejects the file based on strict whitelist validation for the extension and file content.

3. "Write test cases for user roles."

No role types (Admin, Manager, Viewer), no permission matrix, no unauthorized access scenarios, no app domain

Here is a comprehensive set of test cases designed for **User Roles and Permissions (Role-Based Access Control - RBAC)** to ensure proper authorization and access restriction across the system:

1. Positive Test Cases (Authorized Access)

- **TC-POS-01: Admin Access to Admin Dashboard**
 - *Description:* Log in with a user account assigned the Administrator role and navigate to the admin control panel.
 - *Expected Result:* The user is granted full access, and all admin features, settings, and management tools are visible and operational.
- **TC-POS-02: Standard User Access to Profile Settings**
 - *Description:* Log in with a standard User role and navigate to personal account settings.
 - *Expected Result:* The user can view and edit their own profile details, password, and preferences without errors.
- **TC-POS-03: Moderator Access to Content Moderation**
 - *Description:* Log in with a Moderator role and access the content review queue.

- *Expected Result:* The user can view, approve, or flag user-generated content as permitted by their role definition.

2. Negative Test Cases (Unauthorized Access & Privilege Escalation)

- **TC-NEG-01: Standard User Accessing Admin Panel (Direct URL)**
 - *Description:* Log in as a standard User and manually enter the admin URL path (e.g., /admin/settings) in the browser.
 - *Expected Result:* Access is denied. The system redirects the user to an unauthorized error page (403 Forbidden) or the login screen.
- **TC-NEG-02: Guest/Unauthenticated User Accessing Protected Pages**
 - *Description:* Attempt to access a protected internal route (e.g., /dashboard) without logging in.
 - *Expected Result:* The system blocks access and prompts the user to log in.
- **TC-NEG-03: Horizontal Privilege Escalation**
 - *Description:* Log in as User A, then attempt to view or modify User B's private data or profile by altering ID parameters in the API request or URL (e.g., /user/profile?id=102 changed to 103).
 - *Expected Result:* The system blocks the request and returns a 403 Forbidden error, preventing access to another user's data.
- **TC-NEG-04: Hidden Action Execution via API/UI**
 - *Description:* Log in as a read-only viewer and attempt to trigger a "Delete" or "Create" action using developer tools or direct API calls.
 - *Expected Result:* The backend validates the user role, rejects the request, and returns an authorization error.

3. Edge & Boundary Test Cases

- **TC-BND-01: Role Modification During Active Session**
 - *Description:* Log in as a standard user, and while the session is active, an administrator changes their role to an administrator in the database/backend.
 - *Expected Result:* The updated permissions reflect correctly either immediately or upon the next token refresh/page reload (depending on session management architecture).
- **TC-BND-02: Revoking Role Permissions Mid-Session**
 - *Description:* Log in as an administrator, and while logged in, an admin account downgrade occurs or specific permissions are revoked.

- *Expected Result:* Subsequent restricted actions are immediately blocked, invalidating the previous permission state.
- **TC-BND-03: Multi-Role Assignment Handling**
 - *Description:* Assign a user multiple distinct roles simultaneously (e.g., Editor and Support Agent).
 - *Expected Result:* The system correctly aggregates the permissions, granting access to features associated with *both* roles without permission conflicts.

4. Default Role Assignment Test Case

- **TC-DEF-01: New User Registration Default Role**
 - *Description:* Register a new user account through the public sign-up form.
 - *Expected Result:* The user is automatically assigned the correct baseline default role (e.g., Subscriber or Standard User) with no elevated privileges.

ICEPOT reference;; Write test cases for admin role with authorized and unauthorized scenarios for SSO login to MailExchange server

Here is a comprehensive set of test cases tailored for an **Administrator role** performing **Single Sign-On (SSO)** login to a **MailExchange server** environment, covering both authorized access and various unauthorized or restriction scenarios.

1. Positive Test Cases (Authorized Admin SSO Login)

- **TC-AUTH-01: Successful Admin SSO Login (Standard Flow)**
 - **Description:** An authorized administrator navigates to the MailExchange Admin Portal and clicks "Login with SSO".
 - **Steps:**
 1. Open the MailExchange Admin login page.
 2. Click the SSO authentication option and enter the corporate admin email.

3. Complete authentication on the Identity Provider (IdP) portal (e.g., Azure AD, Okta).

- **Expected Result:** The IdP successfully authenticates the user, passes the correct claims/assertions, and redirects the admin to the MailExchange Admin Center with full administrative privileges.

- **TC-AUTH-02: Admin SSO Login with Multi-Factor Authentication (MFA) Enforced**

- **Description:** Verify that an administrator with mandatory MFA requirements completes the full auth loop.

- **Steps:**

1. Initiate admin SSO login.
2. Complete primary credentials on the IdP.
3. Complete the secondary MFA challenge (e.g., authenticator app push, hardware token).

- **Expected Result:** Access is granted only after successful MFA verification, mapping the admin session securely to the MailExchange server.

- **TC-AUTH-03: Session Persistence and Token Renewal**

- **Description:** Verify that an active admin SSO session handles token lifetime and refreshing correctly without forcing unexpected logouts during active administrative tasks.

- **Expected Result:** Session tokens refresh seamlessly via background token exchange until the maximum idle or absolute session timeout is reached.

2. Negative Test Cases (Unauthorized & Restriction Scenarios)

- **TC-UNAUTH-01: Standard User Attempting Admin SSO Access (Role Mismatch)**

- **Description:** A user with a standard "End-User" or "Helpdesk" role in the IdP attempts to log in through the MailExchange Administrator portal.

- **Steps:**

1. Attempt admin SSO login using a non-admin account.
2. Complete IdP authentication.

- **Expected Result:** The MailExchange server rejects the authentication assertion due to missing administrative role claims/permissions, returning a 403 Forbidden or "Unauthorized Access" error page.

- **TC-UNAUTH-02: Disabled or Suspended Admin Account**

- **Description:** An administrator whose account has been disabled or suspended in the central IdP attempts to perform SSO login.

- **Expected Result:** The IdP blocks the authentication request immediately, or if the token is somehow cached, the MailExchange server validates the status against the directory and denies session creation.

- **TC-UNAUTH-03: Invalid or Tampered SAML/OIDC Token Assertion**

- **Description:** Simulate an interception attack where the authentication response (SAML token or OpenID Connect payload) is modified or forged in transit.
- **Expected Result:** The MailExchange server signature validation fails, rejecting the token and terminating the login attempt with a security error log.

- **TC-UNAUTH-04: Cross-Tenant or Unauthorized Domain Access**

- **Description:** An administrator from an external or unverified corporate tenant attempts to authenticate against the primary MailExchange server instance via SSO.
- **Expected Result:** The MailExchange server checks the issuer claim (iss) against the whitelist of trusted tenants and rejects the login attempt.

3. Edge and Security Test Cases

- **TC-SEC-01: Post-Logout Security (Single Logout - SLO)**

- **Description:** Verify that logging out of the MailExchange Admin Portal successfully terminates the session across the IdP and other connected services.
- **Expected Result:** The admin session is destroyed, and clicking the browser's back button does not restore access to administrative functions.

- **TC-SEC-02: Concurrent Admin Session Handling**

- **Description:** An administrator logs into the MailExchange server via SSO from a second browser or device.

- **Expected Result:** Depending on system policy, either the previous session is cleanly terminated or concurrent sessions are logged and tracked securely in the server audit logs.